

AP1
AP2
HACK THE NORTH 2026 · CUSTOM ESP32-C3 FIRMWARE

Badge Finder

Finding a friend with wifi

A

(you)

AP3

B



THE SCENARIO

You're trying to find your friend at Hack the North.

1,000+

hackers in one building

0 bars

of cell signal in the hall

**“by the
tables”**

describes every corner of the
venue

We flashed a custom firmware



your friend is in here. somewhere.

The badge in your hands

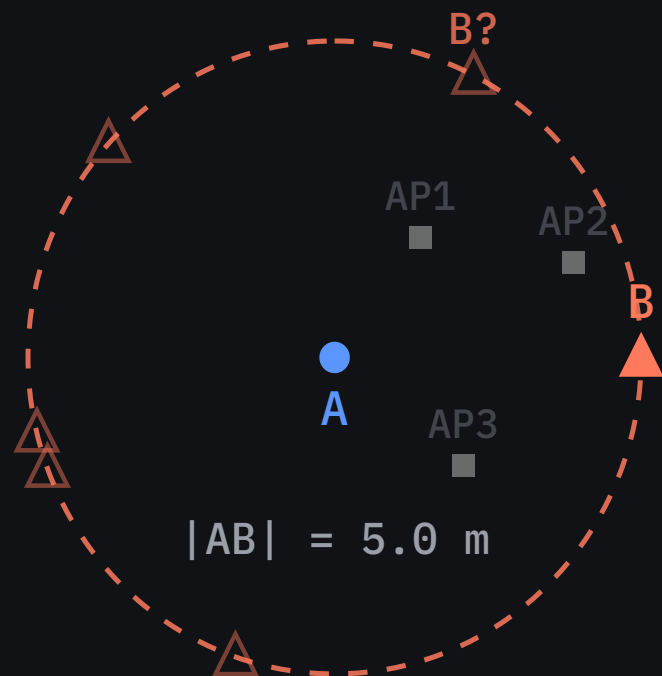
WHAT IT HAS

- WiFi — full scan, ESP-NOW, SoftAP
- 802.11mc FTM — time-of-flight ranging
- BLE — raw advertise + scan
- Accelerometer — step counting
- 6 RGB LEDs + 320×240 display

WHAT IT DOESN'T

- GPS — useless indoors anyway
- Magnetometer — no compass
- Gyroscope — can't track turning
- UWB / AoA antennas — no AirTag tricks

It has no idea which way it is pointing.



$|AB| = 5.0 \text{ m}$ — measured precisely, in every direction at once

WHAT WE CAN MEASURE

A distance is only a ring

FTM gives the distance to B

But it's only a radius. **B is anywhere on the ring.**

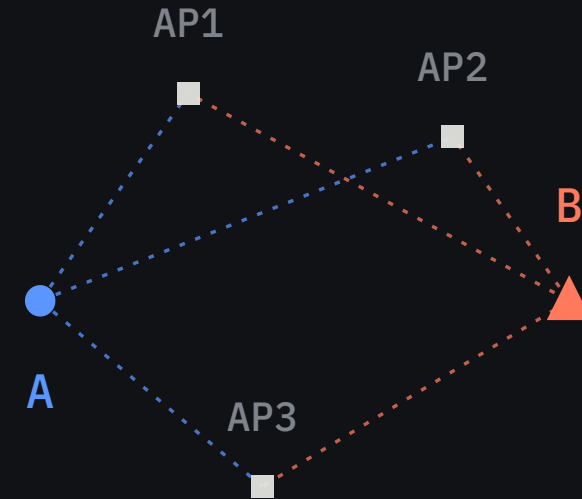
We need a direction.

THE KEY INSIGHT

We are not alone in the room

5–8 access points. They never move. **Both badges** see them.

Free, static reference points.

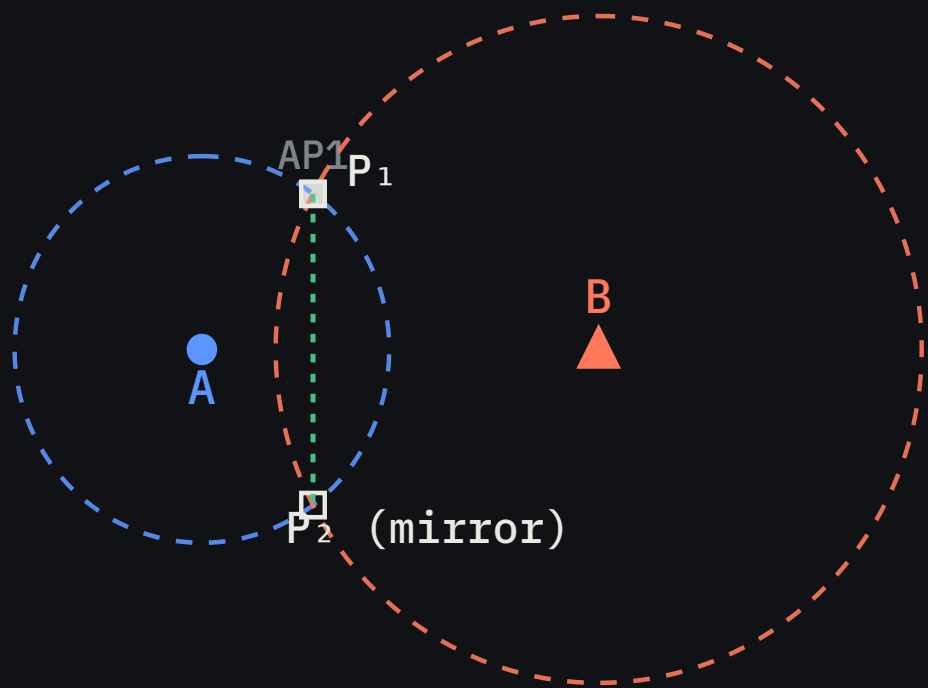


A measures each AP; B does the same and replies over ESP-NOW (16 APs, 114 bytes)

STAGE 1 · STAGE 2

HOW THE MATH WORKS

two captures, one short walk



one shared AP → one circle pair → two candidates

STAGE 1 · FIRST CAPTURE

Two circles, two answers

RSSI → rough distances d_A , d_B . The circles cross at **two mirror points**.

for each shared AP, with A at (0,0) and B at (d,0):

$$x = (d^2 + d_A^2 - d_B^2) / 2d$$

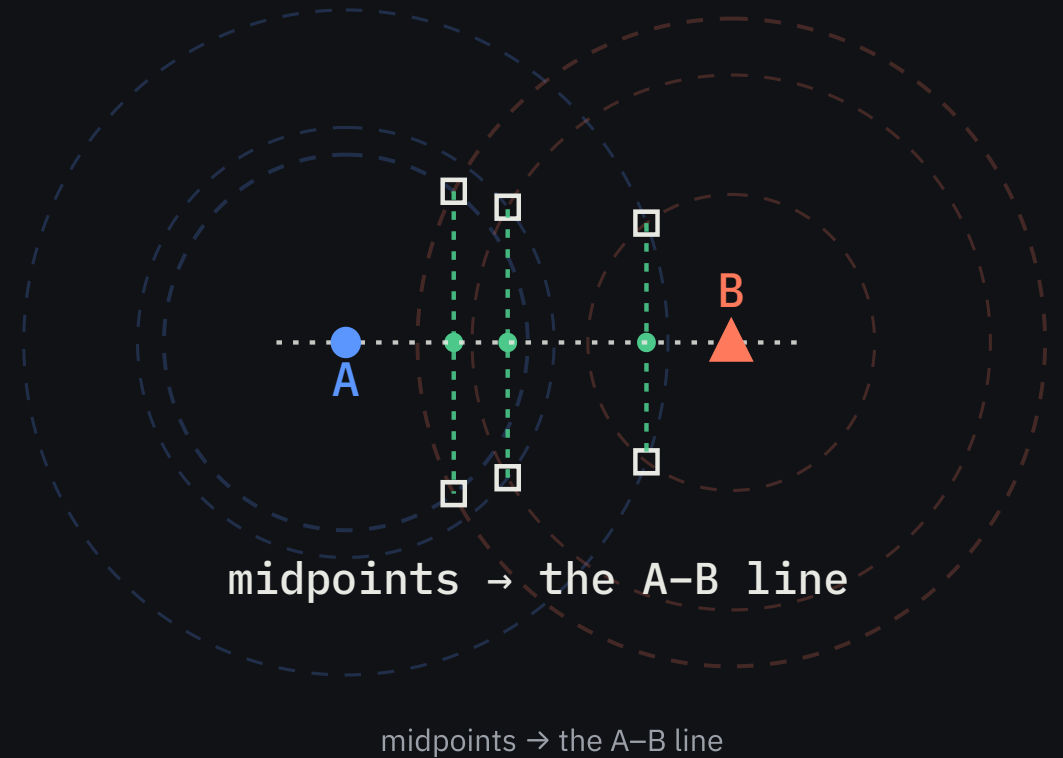
$$y = \pm \sqrt{d_A^2 - x^2}$$

STAGE 1 · GEOMETRY CHECK

The chords hand us the A–B line

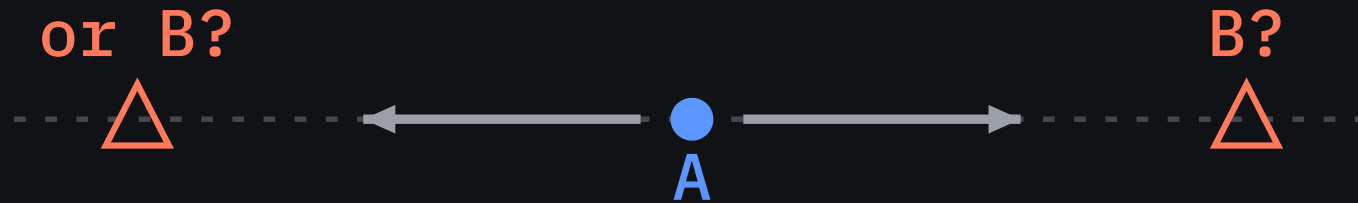
The chords are **all parallel** — 90.00° , every time.

Their midpoints lie on one line: **the A–B line**.

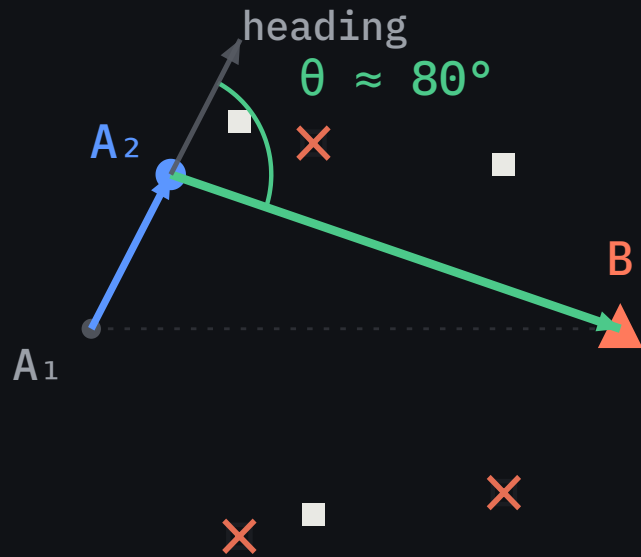


STAGE 1 · THE LIMIT

We know the axis — not the direction



Both directions fit the data — toward B, or exactly away.



three anchors — A_1 , B , A_2 — pin the map to your walk

STAGE 2 · SECOND CAPTURE

Walk a few steps, measure again

Steps count the metres. Direction = “straight ahead”

Re-scan from A_2 solves B — **up to a mirror**. Ranges are mirror-blind: **the spin (or “warmer”) picks the side.**

the on-badge solver (walk direction = $+x$):

$$x = (d_1^2 - d_2^2 + s^2) / 2s$$

$$\theta = \text{atan2}(\sqrt{d_1^2 - x^2}, x)$$

Why this beats spinning in place

$\pm 29^\circ \rightarrow \pm 15-25^\circ$

predicted, vs a single body-shadow spin —
ground-truth run pending

114 bytes

16 APs in one ESP-NOW packet

< 4 m

mean AP residual on the badge $\Rightarrow$ the
geometry is consistent

Running on both badges — the TRIANGULATE page.

Accuracy vs the closing-rate compass: **not measured yet**. That's the next test.

APs are static $\rightarrow$ **cache the map $\rightarrow$ one-scan locates.**

What can go wrong

THE ERROR

30–50%

error in a single WiFi distance estimate

The fix: a rolling average. The noise changes every scan — the truth doesn't.

THE ASSUMPTIONS

- No bodies blocking — a person costs 10–20 dB
- Only light obstruction between badges
- You walk roughly straight
- Hallway-scale, not an AirTag — the screen says “LEFT or RIGHT: walk one way, warmer = correct”

A_2 (you)

B — found



DEMO

live, on two badges

Questions?

Hack the North 2026 badge · custom ESP-IDF firmware · TRIANGULATE page, live on two badges